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MIL-DTL-62265C(AT)
w/AMENDMENT 1
17 December 2003
SUPERSEDING
MIL-PRF-62265C(AT)
30 September 1998

DETAIL SPECIFICATION

FILTER ELEMENT AND RELIEF VALVE, FLUID PRESSURE, HYDRAULIC (M-60 TANK SERIES)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance and manufacture requirements for a hydraulic filter element and relief valve, disposable type, 3 and 5 micron filtration, herein referred to as the assembly.

1.2 Classification. Assemblies will be of the following assembly part numbers (see 6.2):

12257235	- 5 microns filtration, 10 gallons per minute flow rate.
12270316-1	- 3 microns filtration, 12 gallons per minute flow rate.
12270316-2	- 3 microns filtration, 3 gallons per minute flow rate.

Comments, suggestions, or questions on this document should be addressed to Tank-automotive and Armaments Command, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to standardization@tacom.army.mil Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic - Hydrocarbon Base.
MIL-STD-810	- Environmental Test Methods.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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DRAWINGS

ARMY

12257235	- Filter Element and Relief Valve Assembly.
12270316	- Filter Element and Relief Valve Assembly.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for hydraulic valves. The materials shall be capable of meeting all the operational and environmental requirements specified herein (see 4.8). Recovered materials shall be used to the maximum extent practicable.

3.2.1 Metals. All metals used in the construction of the assembly shall be corrosion-resistant or shall be adequately protected to resist corrosion during the normal service life of the filter element (see 4.8).

3.2.2 Dissimilar metals. The use of dissimilar metals in contact with each other shall be avoided. Where contact between dissimilar metals is unavoidable, they shall be suitable protected against electrolytic corrosion (see 4.8).

3.2.3 Filter element material. The material of the filter element shall not change the composition of the hydraulic fluid MIL-H-46170, nor shall the material be altered enough to use or storage to affect the intended function of the filter element (see 4.8).

3.2.4 Hazardous materials. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries (see 4.8).

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3.3 Design and construction. Assembly shall conform to Drawing 12257235 or 12270316, as specified (see 6.2), and other applicable drawings, specifications, and standards. Parts shall be interchangeable as to fit, form, and function with all other units bearing same part number. Assembly shall be interchangeable with accepted assemblies and with current mountings (see 4.7).

3.3.1 Relief valve. The assembly shall incorporate a permanently installed, nonadjustable, by-pass relief valve capable of performance specified in 3.4.9 (see 4.9.9).

3.3.2 Flow direction. Flow through the element shall be from the outside in as verified during testing (see 4.9.4 through 4.9.9).

3.4 Operating requirements.

3.4.1 Bubble point. There shall be no bubbles visible when the filter element is pressurized to 0.1 inch (in.) of water less than the minimum bubble point determined during preproduction testing (see 4.9.1).

3.4.2 Filtration. The filter element shall remove 97 to 99 percent by weight of all contaminants as specified in table I (see 4.9.2).

TABLE I. Contaminants filtration.

Characteristic	Assembly		
	12257235	12270316-1	12270316-2
Filtration (μ) 1/	5	3	3
Flow rate (gal/min) 2/	10	12	3
Pressure drop (psid) 3/	6	12	10

1/ μ = micron
2/ gal/min = gallon per minute
3/ psid = pound per square inch differential

3.4.3 Flow rate. The rated flow capacity of the filter element shall be as specified in table I (see 4.9.3).

3.4.4 Pressure drop. The pressure drop across the clean filter element shall not exceed the amount specified in table I at flow rate specified (see 4.9.4).

3.4.5 Dirt capacity. The filter element shall hold the specified amount of AC Fine Test Dust without exceeding the differential pressure as specified in table II at flow rate specified in 3.4.3 (see 4.9.5).

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TABLE II. Dirt capacity.

Assembly	Incremental additions of contaminant gram (g)	Minimum capacity (g)	Maximum differential pressure (psid)
12257235	2.5	20.0	100
12270316-1	1.2	4.8	45
12270316-2	0.6	2.6	45

3.4.6 Efficiency. The assembly shall remove 97 percent, by weight, of contaminant from a mixture of 2000 mL of clean hydraulic fluid MIL-H-46170 or commercial equivalent (less than 0.0004 g of contaminant per 2000 mL of fluid), and a 5 mL slurry in accordance with table III (see 4.9.6).

TABLE III. Slurry mixture.

Assembly	A/C fine test dust (g)	APM F-9 beads (g)
12257235	0.15	0.15
12270316-1	0.25	0.25
12270316-2	0.15	0.15

The efficiency shall be determined by the following formula:

$$\text{Efficiency} = \frac{A-(B-C)}{A} \times 100$$

Where: A = (Add value) = Amount of test contaminant passed through system when there is no filter element in the filter housing.

B = (Contaminant value) = Amount of test contaminant passed through the test filter assembly with the filter element installed.

C = (Blank value) = Amount of contaminant attributed to the test system and test filter assembly when no contaminant has been added. This amount shall be less than 0.0007 g.

3.4.7 Collapse pressure. The filter element shall withstand the minimum differential pressure, at flow rate specified in table IV without inward structural failure or filter medial failure. The differential pressure shall be obtained by adding AC Fine Test Dust to the test system

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while maintaining flow rate. The bubble point shall not be below the minimum established during preproduction or initial production testing (see 4.9.7).

TABLE IV. Collapse pressure.

Assembly	Differential pressure (pounds per square inch) (psi)	Rated flow (gal/min)
12257235	250	10
12270316-1	150	12
12270316-2	150	3

3.4.8 Flow fatigue. The filter element shall be capable of withstanding flow fatigue cycling specified in table V without evidence of structural failure. The bubble point shall not be less than the minimum value established during preproduction or initial production testing (see 4.9.8).

TABLE V. Flow fatigue.

	Pressure drop (pounds per square inch gage) (psig)	
Number cycles	12257235	12270316-1, -2
225 000	20	20
22 500	50	50
2 000	100	75
500	150	100

3.4.9 Relief valve performance.

3.4.9.1 Cracking pressure. The relief valve shall crack at the differential pressure specified in table VI. Cracking pressure is defined as that pressure at which leakage through the valve exceeds that specified in table VI (see 4.9.9.1).

TABLE VI. Relief valve.

Assembly	Flow capacity		Cracking		Reseat	
	Rate (gal/min)	Differential pressure (psi)	Differential pressure (psi)	Leakage cubic centimeters per min (cm^2/min)	Differential pressure (psi)	Leakage (cm^2/min)
12257235	15.5	190	125 ±25	16	60	8
12270316-1	18	120	50 ±5	4	30	2
12270316-2	4.5	120	50 ±5	4	30	2

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3.4.9.2 Flow capacity. With the flow specified in table VI through the relief valve, the differential pressure across the relief valve shall not exceed the pressure specified in table IV (see 4.9.9.2).

3.4.9.3 Reseat pressure. The relief valve shall reseat at the differential pressure as specified in table VI. Reseat pressure is defined as that pressure at which leakage through the valve does not exceed the amount specified in table VI (see 4.9.9.3).

3.4.9.4 Endurance. The relief valve shall be capable of withstanding 2000 cycles of opening and closing the valve from zero to the flow capacity specified in table VI (see 4.9.9.4).

3.5 Interface requirements.

3.5.1 Dimensional constraints. Dimensional constraints of the assembly shall be as specified on Drawing 12257235 or 12270316, and drawings referenced therein (see 4.10.1).

3.5.2 Weight. The weight of the assembly shall not exceed 1.5 pounds (lb) (see Drawing 12257235) or 0.8 lb (see Drawing 12270316) (see 4.10.2).

3.6 Support and ownership requirements.

3.6.1 Identification marking. Identification marking shall be permanent and legible and shall include, as a minimum, the manufacturer's name and part number, the part or identifying number (PIN) and the National Stock Number (NSN). Special markings, if any (see 6.2), shall be as specified (see 4.11.1).

3.6.2 Workmanship. The assembly shall be free of defects detrimental to its intended use and service life (see 4.11.2).

3.7 Environmental requirements. The assembly shall meet the requirements of 3.4 after being subjected to the environmental conditions specified below.

3.7.1 Temperature. The assembly shall be capable of operating at rated flow over the temperature range of -65°F to +275°F, and shall not be damaged when subjected to storage conditions of -65°F. In addition, the assembly shall be capable of withstanding 10 flow cycles of cold start at -65°F. The bubble point shall not be less than the minimum value established during first article testing (see 4.12.1).

3.7.2 Vibration. The assembly shall be capable of withstanding the vibration profile per figure 1 for a period of 80 minutes (including up to 4 resonance dwells of 13-1/3 minutes each) in each direction of the 3 mutually perpendicular axes. The filter element media migration shall not exceed 0.004 g (see 4.12.2).

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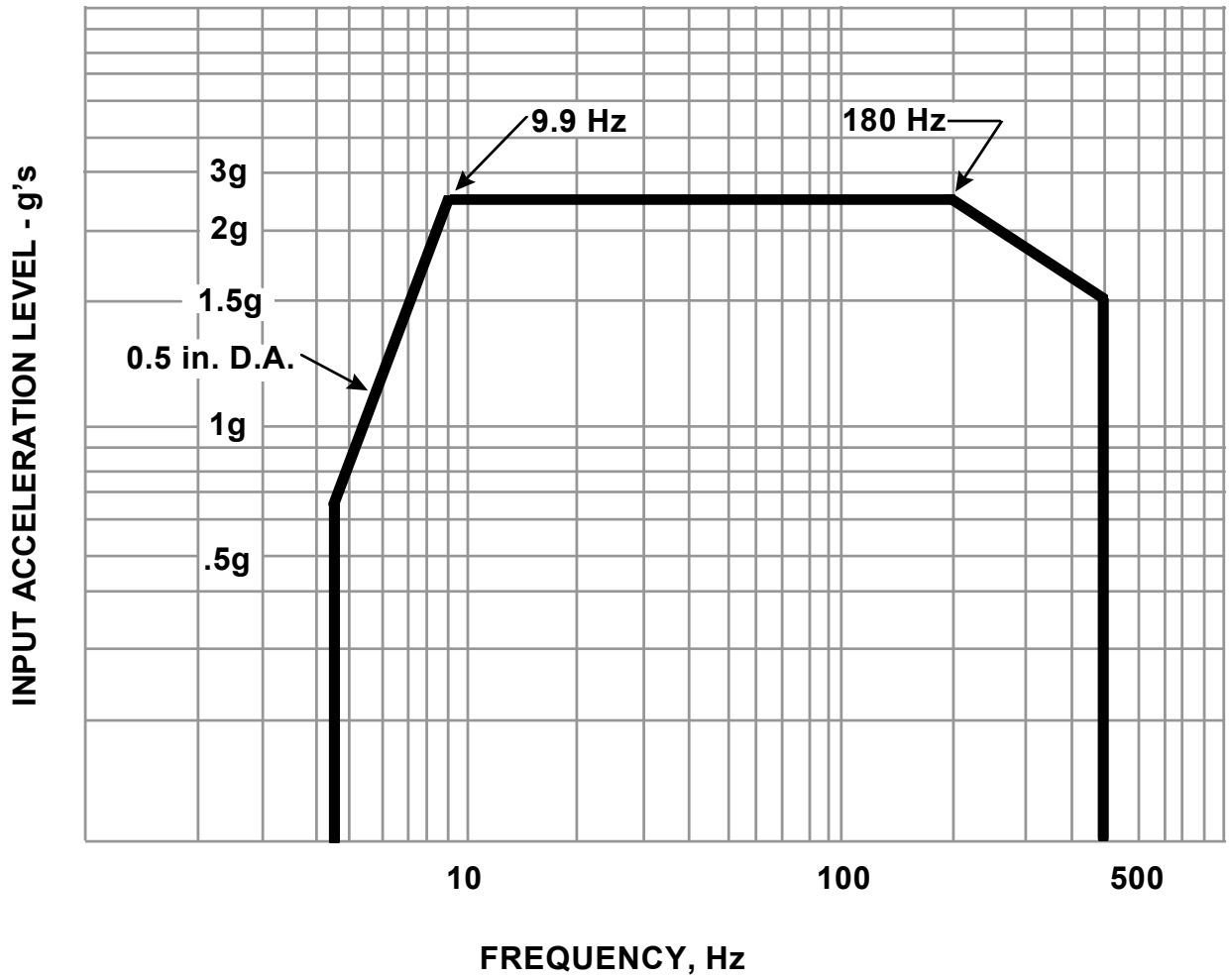


FIGURE 1. Vibration profile.

3.7.3 Shock. The assembly shall be capable of withstanding exposure to three half sine wave shock pulses of 40 ± 4 g for a duration of 18 ± 3 milliseconds (ms) applied in both directions along three mutually perpendicular axes (see 4.12.3).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

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4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 and 4.6.2 and the tests of 4.9.1.1 through 4.12.3 (see table VII).

TABLE VII. Classification of inspections.

Title	Require-ment	Method	First article								Conform-ance	Examin-ation
			Preproduction				Initial Production					
			1	2	3	4	1	2	3	4		
Materials	3.2	4.8	X	X	X		X	X	X	X		
Metals	3.2.1	4.8										
Dissimilar metals	3.2.2	4.8										
Filter element material	3.2.3	4.8										
Hazardous materials	3.2.4	4.8										
Design and construction	3.3	4.7										
Relief valve	3.3.1	4.9.9										
Flow direction	3.3.2	4.9.4 thru 4.9.9										
Operating requirements	3.4											
Bubble point	3.4.1	4.9.1										
Bubble point - alcohol	3.4.1	4.9.1.1	X	X	X	X	X	X	X	X		
Bubble point - hydraulic fluid	3.4.1	4.9.1.2	X	X	X	X	X	X	X	X		
Bubble point - alcohol	3.4.1	4.9.1.3				X				X	X	
Filtration	3.4.2	4.9.2	X				X					
Flow rate	3.4.3	4.9.3				X				X		
Pressure drop	3.4.4	4.9.4				X				X		
Dirt capacity	3.4.5	4.9.5				X				X		
Efficiency	3.4.6	4.9.6	X				X					
Collapse pressure	3.4.7	4.9.7				X				X		

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TABLE VII. Classification of inspections - Continued.

Title	Require-ment	Method	First article								Conform-ance	Examin-ation
			Preproduction				Initial Production					
			1	2	3	4	1	2	3	4		
Flow fatigue	3.4.8	4.9.8		X				X				
Relief valve	3.4.9	4.9.9										
Cracking pressure	3.4.9.1	4.9.9.1			X				X			
Flow capacity	3.4.9.2	4.9.9.2			X				X			
Reseat pressure	3.4.9.3	4.9.9.3			X				X			
Endurance	3.4.9.4	4.9.9.4	X	X	X	X	X	X	X	X		
Interface requirements	3.5	4.10										
Dimensional constraints	3.5.1	4.10.1										X
Weight	3.5.2	4.10.2	X	X	X		X	X	X	X		X
Support and ownership requirements	3.6	4.11										
Identification marking	3.6.1	4.11.1										X
Workmanship	3.6.2	4.11.2										X
Operating environmental requirements	3.7	4.12										
Low temperature	3.7.1	4.12.1.1	X	X	X	X	X	X	X	X		
Vibration	3.7.2	4.12.2	X	X	X	X	X	X	X	X		
Shock	3.7.3	4.12.3			X				X			

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction shall be performed on four valve assemblies.

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article initial production shall be performed on four valve assemblies randomly selected from the first ten production valves produced.

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4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests of 4.9.1.3 (see table VII).

4.4 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with the contract order.

4.5 Examination. Each valve assembly shall be examined for compliance with the requirements specified in 3.3, 3.5.1, 3.6.1, and 3.6.2. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification follow in- failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Acceptance tests (100 percent). Each valve assembly shall be subjected to the test specified in 4.9.1.3.

4.6.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Temperature: 73° ± 18°F
b. Pressure: 24 to 30.5 in. of mercury (Hg)
c. Humidity: 50 ± 30 percent (%)
d. Hydraulic fluid temperature: 100° ± 10°F

4.6.2 Methods of inspection. Filter shall be inspected in accordance with the following requirements:

a. A filter head assembly suitable for mounting and providing fluid flow through the filter element and relief valve (assembly) shall be provided for testing as required.
b. Unless otherwise specified, the hydraulic fluid used for all tests shall conform to MIL-H-46170 or commercial equivalent and shall be continuously filtered through a 5 micron absolute filter for Drawing 12257235 assembly and a 3 micron absolute filter for Drawing 12270316 assembly. The contamination level shall not exceed 0.0004g per 2000 mL of fluid.
c. Unless otherwise specified, the air used to pressurize the test apparatus shall be filtered through a 0.45 µ membrane filter (type HA Millipore, or equivalent).

4.7 Verification methods. Acceptable verification methods included in this section are visual inspection and measurement, sample tests, full-scale demonstration tests, simulation,

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modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.7.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.8 Material. Conformance to 3.2 thru 3.2.4, shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.9 Operating requirements verification.

4.9.1 Bubble point (see 3.4.l).

4.9.1.1 Alcohol, first article testing. To determine the bubble point for the assembly the following method shall be used:

a. Install the assembly in a test fixture which will permit simultaneous pressurization and 360° rotation of the assembly around its longitudinal axis.
b. Cover the assembly to a depth of 0.5 in. with isopropyl alcohol, or equivalent, at 70° ± 5°F. The bubble point test shall be performed within 10 minutes after immersion of the assembly.
c. Connect an air pressure source to the assembly and raise the pressure in small increments.
d. Rotate the assembly slowly during each pressure increment so that the entire filter surface area can be observed for the appearance of the first bubble.
e. The manometer reading (in in. of water) at the time the first bubble is emitted from the assembly is the alcohol bubble point of the element.
f. Inspect the results.

4.9.1.2 Hydraulic fluid, first article testing, and control level establishment.

a. Install the assembly in a test fixture which will permit simultaneous pressurization and 360° rotation of the assembly around its longitudinal axis.
b. Cover the assembly to a depth of 0.5 in. with test fluid MIL-H-46170 or commercial equivalent, at 700° ± 50°F. The bubble point test shall be performed within 10 minutes after immersion of the assembly.
c. Connect an air pressure source to the assembly and raise the pressure in small increments.

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d. Rotate the assembly slowly during each pressure increment so that the entire filter surface area can be observed for the appearance of the first bubble.
e. The manometer reading (in in. of water) at the time the first bubble is emitted from the assembly is the bubble point of the element.
f. Inspect the results.
g. The hydraulic fluid bubble point valve established per steps (a) through (f) will be the base value used for the remainder of tests required under preproduction and initial production testing.

4.9.1.3 Alcohol, conformance tests. The assembly shall be subjected to and pass the following test:

a. Install the assembly in test fixture which will permit simultaneous pressurization and 360° rotation of the assembly around its longitudinal axis.
b. Cover the assembly to a depth of 0.5 in. with isopropyl alcohol, or equivalent, at 70° ± 5°F.
c. Connect an air pressure source to the assembly and slowly pressurize the assembly to a pressure of 0.1 in. of water less than the minimum alcohol bubble point determined during preproduction testing.
d. Rotate the assembly slowly during pressurization in order to observe the entire filter for bubbles. No bubbles are allowed.

4.9.2 Filtration capability. The filtration capability shall be determined by subjecting the assembly to and passing the test of 4.9.6, steps (a) through (g) with the addition of the following (see 3.4.2):

a. All hydraulic fluid passed through the test assembly shall be filtered through a 0.8 µ membrane filter disk AA millipore, or equivalent, into a 5000 mL flask. The 5000 mL flask shall then be washed with 1000 mL of prefiltered petroleum ether or equivalent. The cleaning fluid from the washed flask shall also be passed through the membrane filter disk.
b. The entire membrane filter disk shall be scanned and the largest bead present shall be smaller than 5 µ for Drawing 12257235 assembly, and 3 microns for Drawing 12270316 assembly.

4.9.3 Flow rate determination. The flow rate for the assembly shall be determined in conjunction with 4.9.6 and shall be as specified in table I (see 3.4.3).

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4.9.4 Pressure drop test. To determine conformance compliance to the pressure drop across a clean filter element, the following procedure, or equivalent, shall be used with the relief valve blocked. Hydraulic pressure, adjusted to provide rated flow, shall be applied to the filter element. Pressure shall be monitored on the inlet and outlet sides of the filter. The pressure drop across the filter element shall be the differences between the inlet and outlet pressure and shall not exceed that specified in table I.

4.9.5 Dirt capacity test. The hydraulic pressure shall be adjusted to provide rated flow in accordance with 3.4.3. AC Fine Test Dust shall be injected into the hydraulic fluid at 4 minute intervals in accordance with table II. The differential pressure across the filter element shall be monitored, and shall not exceed the value specified in table II.

4.9.6 Efficiency test. To determine the efficiency of the filter element, the following procedure, or equivalent, shall be used:

a. Install the test sample into the test setup (see figure 2) with a contaminant level of less than 0.0004 g per 2000 mL of fluid.
b. Temperature of the hydraulic fluid shall be 73° ± 18°F.
c. Add a 5 mL slurry containing contaminants specified in table III into the mixing chamber, and agitate for a minimum of 3 minutes.
d. Apply air pressure to force the fluid containing contaminant through the sample filter assembly and out of the sampling port at rated flow.
e. Collect the filtrate in a 5000 mL flask.
f. Using a suitable wash bottle, wash 750 mL of prefiltered petroleum ether or equivalent through the contaminant mixing chamber and the test filter.
g. Collect the fluid in the same flask as the filtrate.
h. A membrane filter 1.85 in. diameter disk, absolute 0.8 µ type AA millipore, or equivalent, shall be washed with 200 mL of prefiltered petroleum ether or equivalent, and dried to constant weight at 125°F. The weight shall be accurate to 0.1 milligram. The disk shall be heated at 125°F for 30 minutes and then cooled 30 minutes in a desiccator.
i. The weighed filter membrane shall be assembled into a suitable filter holder assembly and fitted to a 5000 mL vacuum flask connected to a suitable vacuum (minimum 25 in. of mercury (Hg) vacuum).
j. All collected hydraulic fluid (step g) shall be filtered through the membrane filter. The 5000 mL flask shall then be washed with 1000 mL of prefiltered petroleum ether or equivalent. The cleaning fluids from the washed flask shall also be passed through the membrane filter.
k. Maintaining vacuum, the rim of the membrane filter shall be gently washed with prefiltered petroleum ether or equivalent to remove traces of hydraulic fluid, being very careful not to disturb the cake.

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l. The vacuum shall be shut off and the membrane filter removed. It shall be dried to a constant weight at 125°F, and weighed to 0.1 of a milligram.
m. The difference between the membrane filter weights (steps (h) and (l)) shall be determined.
n. Filtration efficiency shall be determined in accordance with the formula specified in 3.4.6 and shall be no less than 97 percent. The difference in weights as determined in step (m) shall be the contaminant value (B), and the contaminant add value (as appropriate) in step (c) shall be the add value (A) as specified in the formula in 3.4.6.

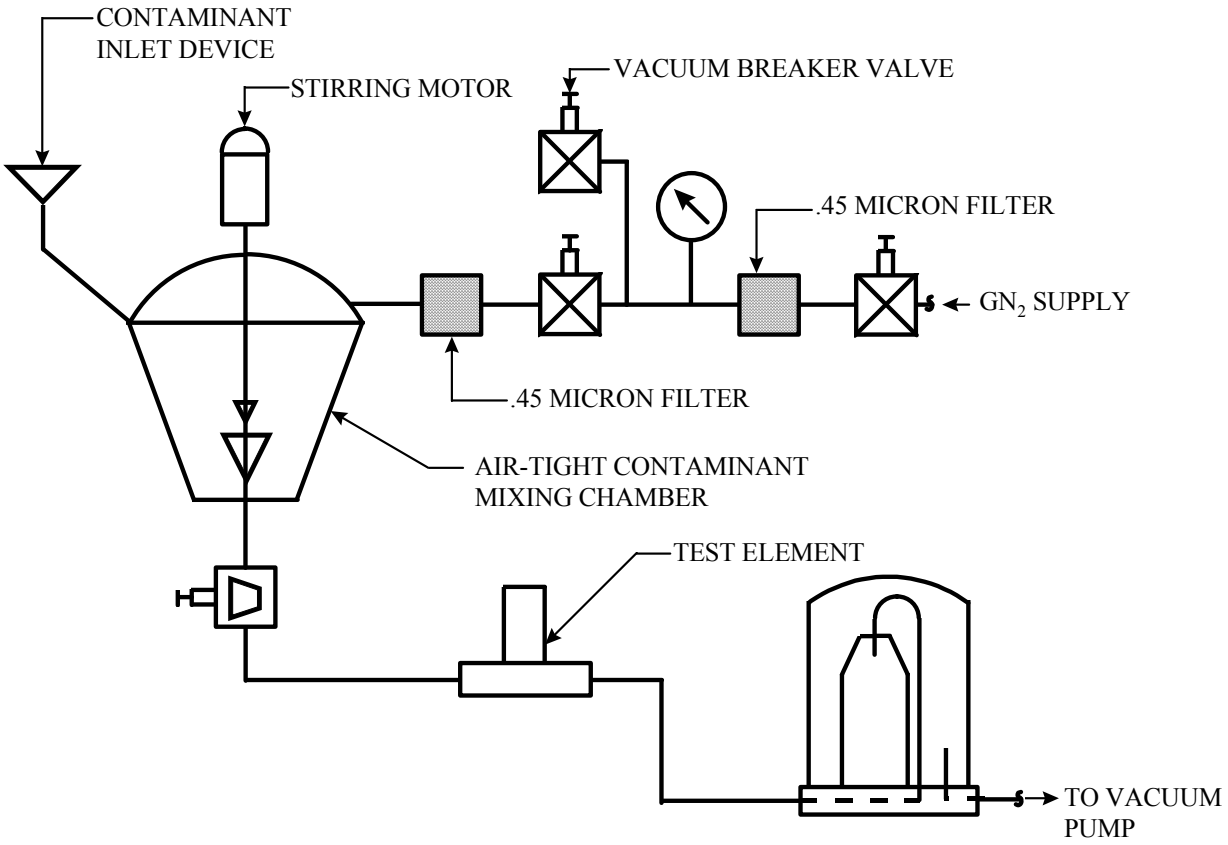


FIGURE 2. Filtration efficiency test setup.

4.9.7 Collapse pressure test. The hydraulic pressure, adjusted to provide rate flow, shall be applied to the filter element. The filter element relief valve shall be blocked. AC Fine Dust shall be injected into the hydraulic fluid until the differential pressure of table IV across the filter element is obtained. Subsequent to the test, the filter element shall be inspected for indications of collapse and subjected to and pass the bubble point test of 4.9.1.2. The bubble point value

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obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing.

4.9.8 Flow fatigue test. The assembly shall be blocked and the temperature of the hydraulic fluid shall be 275°F minimum. The assembly shall be subjected to pressure flow cycles with the fluid flow being increased to a minimum of 30 percent of rated flow, and then shall be decreased to zero flow to complete one cycle. A cycle shall consist of increasing the differential pressure across the assembly from zero the maximum specified and back to zero by first increasing and then decreasing the flow through the filter element. The filter element shall be progressively loaded with fine dust (AC), or equivalent, to raise the differential pressure (pressure drop) as specified in table V. Peak pressure shall be reached within 25 percent of the cycle and held for at least 50 percent of the cycle. The cycling rate shall be 100 cpm maximum and the number of cycles at each differential pressure shall be in accordance with table V. Upon completion of this test, the filter element shall be bubble point tested in accordance with 4.9.1.2. The value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing as in 4.9.1.2.

4.9.9 Relief valve testing (see 3.4.9).

4.9.9.1 Cracking pressure determination. The filter element media shall be blocked and the assembly installed in a suitable housing. Pressure shall be applied to the inlet port of the filter housing beginning with a pressure of 40 psi and increasing in increments of 20 psi, or less, until cracking pressure of the relief valve in the assembly being tested is reached. At each increment, the pressure shall remain constant for 5 minutes. At each pressure increment, valve opening shall not be considered until the beginning of the third minute. The leakage rate during the last 3 minutes of each pressure increment shall be noted. Cracking pressure shall be considered as the pressure at which the leakage rate through the relief valve amounts to the value specified in table VI.

4.9.9.2 Flow capacity determination. This test shall be conducted in conjunction with 4.9.9.1. Pressure shall be increased beyond the cracking pressure, until the flow rate of table VI through the relief valve is obtained. A pressure differential between the inlet and outlet ports shall not exceed the pressure specified in table VI.

4.9.9.3 Reseat pressure determination. This test shall be conducted in conjunction with 4.9.9.1. The pressure shall be reduced in 10 psi increments until leakage through the relief valve does not exceed the rate specified in table VI. The resulting pressure shall be considered the relief valve reseating pressure and shall occur at a pressure of not less than that specified in table VI.

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4.9.9.4 Endurance test. This test shall be conducted in conjunction with 4.9.9.1. The relief valve shall be cycled 2000 times from zero flow to the flow specified in table VI, and back to zero flow. At the conclusion of 2000 cycles, the cracking and reseating pressure of 4.9.9.1 and 4.9.9.3 shall be verified.

4.10 Interface requirements verification.

4.10.1 Dimensional constraints. To determine the dimensional constraints, measuring equipment shall be used (see 4.7). Dimensions not within tolerance shall cause for rejection.

4.10.2 Weight. To determine the weight of the filter element and relief valves, a scale shall be used (see 4.7). Weight exceeding the values specified in 3.5.2 shall cause for rejection.

4.11 Support and ownership requirement verification.

4.11.1 Identification marking. To determine the legibility and permanence of the marking, a visual examination shall be used (see 4.5 and 4.7). Incorrect or illegible marking shall cause for rejection.

4.11.2 Workmanship. To determine workmanship of the assembly, a visual and mechanical examination shall be used to find detrimental defects of the filter element and relief valve (see 4.5 and 4.7).

4.12 Environmental requirements verification. Assemblies undergoing preproduction, initial production, or control testing shall be subjected to hydraulic fluid bubble point tests (see 4.9.1.2) before and after each of the environmental tests performed. Assemblies undergoing environmental control test shall have passed the acceptance tests (see 3.7).

4.12.1 Temperature (see 3.7.1).

4.12.1.1 Low temperature. The assembly shall be subjected to low temperature testing at -65 °F for 12 hours in accordance with procedure I, method 502 of MIL-STD-810, or equivalent (see 4.7). While stabilized at -65 °F, the assembly shall be subjected to a total of 10 flow cycles at a differential pressure across the assembly from zero to 90 psi, and back to zero by first increasing, then decreasing, the flow through the assembly. Each flow cycle shall be 15 ± 5 seconds in duration. The test fluid and assembly shall be maintained at -65 °F or less for the duration of the test. The flow shall not exceed 150 percent of the rated flow and shall be at a minimum of 0.5 gal/min. Subsequent to low pressure exposure, perform the hydraulic fluid bubble point test of 4.9.1.2. The bubble point value obtained shall not be less than the minimum value established during preproduction or initial production testing of 4.9.1.2.

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4.12.1.2 High temperature test. This test shall be performed in conjunction with the flow fatigue test of 4.9.8.

4.12.2 Vibration test. The assembly shall be filled with prefiltered hydraulic fluid and the inlet and outlet ports shall be capped with precleaned caps. The assembly shall be subjected to and pass the test using procedure VIII, method 514 of MIL-STD-810, or equivalent (see 4.7), with the following conditions.

a. Test level shall be the vibration curve shown in figure 1. Time schedule shall be a total of 80 minutes per axis including 13-1/3 minutes dwell time at each resonance.
b. Connections and instrumentation shall be attached to the assembly to permit cycling as specified in (c) through (d).
c. Prior to vibration test, the housing shall be flushed with 2000 mL of prefiltered hydraulic oil, and effluent contamination level shall be determined. The contaminant value shall represent the blank value "C", as specified in the migration formula below.
d. Conduct resonance search along each axis at reduced vibration amplitude over the frequency range of 5-500 Hz. Up to four (4) resonant frequencies shall be identified (the four most severe resonant frequencies if more than four present) in each axis for the resonance dwell tests.
e. The assembly unit shall be subjected to resonance dwell and frequency cycling tests with test levels and time schedule as identified in "a" above.
f. Subsequent to vibration tests, the caps shall be removed from the ports, and the assembly shall be flushed with 2000 mL of prefiltered hydraulic oil. The effluent shall be collected and the contamination level shall be determined. The contaminant value shall represent the analysis value "A" as specified in the following migration formula:

   X = A-C
   X = Migration analysis value
   A = Analysis value
   C = Blank value (per step c)

   The filter element media migration shall not exceed 0.004 g.
g. Perform the hydraulic bubble point test on the flushed assembly in accordance with 4.9.1.2. The bubble point value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing.

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4.12.3 Shock test. The assembly shall be installed in a test fixture and subjected to shock to pulses of 40 g for a duration of 18 ± 3 ms per MIL-STD-810, method 514, procedure I, or equivalent (see 4.7). Three (3) half-sine-wave shock pulses shall be applied to the assembly in each direction of 3 mutually perpendicular axes. After the test, the assembly shall be examined for evidence of damage and subjected to the hydraulic fluid bubble point test of 4.9.1.2. The bubble point value obtained shall not be less than the minimum value established during hydraulic fluid preproduction or initial production testing as in 4.9.1.2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use, The filter element and relief valve assembly covered by this specification are of the disposable type, 3 and 5 µ filtration. They are military unique because of the special design, dimensional constraints, and their intended use in the hydraulic systems of M60A1, M60A2, and M60A3 tanks operating with hydraulic fluid conforming to MIL-H-46170 and have no commercial applications.

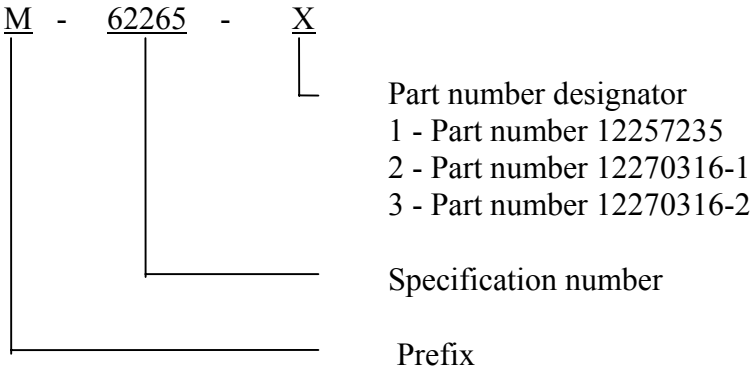
6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

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c. If first article assembly is required (see 3.1).
d. Assembly drawing number, part number, and PIN (see 3.3, 3.6.1 and 6.3).
e. Any special marking, if required (see 3.6.1).
f. If first article preproduction inspection is required (see 4.2.1).
g. If initial production inspection is other than as specified (see 4.2.2).
h. If inspection conditions are other than as specified (see 4.6.1).
i. Packaging requirements (see 5.1).

6.3 Part of Identifying Number (PIN). The PINs to be used for assemblies acquired to this specification are created as follows:



6.4 First article preproduction. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applies, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table VII and their designated frequency based on a risk assessment for the procurement.

6.6 Subject term (key word) listings.

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Bubble point
Combat vehicle, tracked
Flow capacity

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 4820-0922)

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